RFM segmentation of bank clients

Recency · Frequency · Monetary

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Problem

Marketing gets a flat customer list and targets everyone the same way.

- No per-customer segment label
- Duplicated ad-hoc segment definitions
- No way to measure segment-level lift

Goal: give marketing a targetable segment per customer.

Data

Synthetic transaction log (`generate_data.py` , seed 42).

Period	2023-01-01 → 2024-03-30
Volume	10 000 txns → 9 492 after anomaly drop → **1 983** active clients
Schema	`customer_id, transaction_date, amount, product_category, payment_method, is_anomaly`

⚠️ Synthetic — segment sizes reflect the generator, not a real bank.

Method

1. **Load & clean** — drop `is_anomaly == 1`
2. **RFM table** — per customer: recency (days), frequency (count), monetary (sum)
3. **Quartile score 1–4** — higher = better; `RFMClass` = 3-digit combo
4. **Segment** — combo-based (not sum-based), so `444` ≠ `233`

```
R≥4 & F≥4 & M≥4 → Champions
R≥3 & F≥3 & M≥3 → Loyal
R≥3 & F≥2 & M≥3 → Potential Loyalists
R≤2 & F≥3       → At Risk
R≤2 & F≤2       → Hibernating
else            → Need Attention
```

Results — segments

Segment	Customers	Avg recency	Avg freq	Avg monetary	Avg RFM
Champions	126	11.35	8.16	27 608.83	12.00
Loyal	376	32.13	6.40	20 692.31	10.32
Potential Loyalists	79	32.62	3.97	16 777.17	8.57
At Risk	349	123.85	6.30	19 381.56	8.37
Need Attention	410	30.39	4.07	8 962.92	7.20
Hibernating	643	177.47	2.92	8 786.79	4.56

Results — revenue concentration

Gini = **0.324** — monetary value is moderately concentrated; the top 20% of customers hold 38.2% of revenue.

- **Champions** (6.4% of base) → 12.1% of revenue, concentration ratio **1.91**
- **At Risk** (17.6%) → 23.6% of revenue, ratio **1.34** — the highest-leverage win-back pool
- **Potential Loyalists** (4.0%) → 4.6% of revenue, ratio **1.16**
- **Hibernating** (32.4% of base) → 19.7% of revenue, ratio **0.61** — under-indexes now
- **Need Attention** (20.7%) → 12.8% of revenue, ratio **0.62**

→ Invest in Champions/Loyal/At Risk; cheap win-back for Hibernating; don't over-spend on Need Attention.

Hypotheses

#	Hypothesis	Test	Result
H1	Higher Monetary ↔ more distinct products	Spearman ρ	**ρ=0.61, p<0.001** ✅ (Holm 0.000)
H5	Mobile-bank users have higher Monetary	Mann-Whitney U (two-sided)	**med 11 840 vs 13 445, p=0.0003 — significant but OPPOSITE direction** ❌
H6	Segment × product-category associated	χ^2 (customer level), dof 25	**χ^2=52.3, Cramér's V=0.073, p=0.002** ✅ (Holm 0.006)
H7	Mobile users have lower Recency	Mann-Whitney U (two-sided)	**med 69 vs 66, p=0.43** ❌ (Holm 0.80)

Recommendations

- **Champions (126, 6.4%):** retention rewards + referral asks.
- **At Risk (349, 17.6%):** re-engagement campaign before they slide to Hibernating — see [EXP-001].
- **Hibernating (643, 32.4%):** cheap win-back emails, don't over-invest.
- **Need Attention (410, 20.7%):** upsell to lift M, or silent churn risk.
- **Potential Loyalists (79, 4.0%):** highest AOV — upsell premium products to grow F.

What's in the repo

```
rfm_analysis.py        pipeline: load → score → segment → export
generate_data.py       synthetic dataset (seed 42)
hypotheses.py          5 testable + 3 untestable checks
dashboard/app.py       Streamlit dashboard (5 tabs)
docs/                  data dictionary, SQL templates, A/B design, PRD, metrics
tests/                 54 tests (pytest)
presentation/          this deck (Marp)
```

`uv sync --all-extras` · `pytest` · `streamlit run dashboard/app.py`

Thank you

Questions?

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